# Precalculus — Algebra I

Version 4 $- \epsilon$

by

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$\sigma o \phi \acute{\iota} \alpha$

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# Table of Contents

**Preface** v

**1 Introduction to Functions** 1

1.1 Functions and their Representations . . . 1

1.2 Constant and Linear Functions . . . . . 2

1.2.1 Constant Functions . . . . . 2

1.2.2 Linear Functions . . . . . 10

1.2.3 Linear Regression . . . . . 33

1.2.4 The Average Rate of Change of a Function . . . . . 45

1.2.5 Exercises . . . . . 54

1.2.6 Answers . . . . . 72

1.3 Absolute Value Functions . . . . . 86

1.4 Quadratic Functions . . . . . 87

# Preface

Thank you for your interest in our book, but more importantly, thank you for taking the time to read the Preface. I always read the Prefaces of the textbooks which I use in my classes because I believe it is in the Preface where I begin to understand the authors - who they are, what their motivation for writing the book was, and what they hope the reader will get out of reading the text. Pedagogical issues such as content organization and how professors and students should best use a book can usually be gleaned out of its Table of Contents, but the reasons behind the choices authors make should be shared in the Preface. Also, I feel that the Preface of a textbook should demonstrate the authors' love of their discipline and passion for teaching, so that I come away believing that they really want to help students and not just make money. Thus, I thank my fel-

low Preface-readers again for giving me the opportunity to share with you the need and vision which guided the creation of this book and passion which both Carl and I hold for Mathematics and the teaching of it.

Carl and I are natives of Northeast Ohio. We met in graduate school at Kent State University in 1997. I finished my Ph.D in Pure Mathematics in August 1998 and started teaching at Lorain County Community College in Elyria, Ohio just two days after graduation. Carl earned his Ph.D in Pure Mathematics in August 2000 and started teaching at Lakeland Community College in Kirtland, Ohio that same month. Our schools are fairly similar in size and mission and each serves a similar population of students. The students range in age from about 16 (Ohio has a Post-Secondary Enrollment Option program which allows high school students to take college courses for free while still in high school.) to over 65. Many of the "non-traditional" students are returning to school in order to change careers. A majority of the students at both schools receive some sort of financial aid, be it scholarships from the schools' foundations, state-funded grants or federal financial aid like student loans, and many of them have lives busied by family and job demands. Some will be taking their Associate degrees and entering (or re-entering) the workforce while others will be continuing on to a four-year college or uni-

versity. Despite their many differences, our students share one common attribute: they do not want to spend $200 on a College Algebra book.

The challenge of reducing the cost of textbooks is one that many states, including Ohio, are taking quite seriously. Indeed, state-level leaders have started to work with faculty from several of the colleges and universities in Ohio and with the major publishers as well. That process will take considerable time so Carl and I came up with a plan of our own. We decided that the best way to help our students right now was to write our own College Algebra book and give it away electronically for free. We were granted sabbaticals from our respective institutions for the Spring semester of 2009 and actually began writing the textbook on December 16, 2008. Using an open-source text editor called TexNicCenter and an open-source distribution of LaTeX called MikTex 2.7, Carl and I wrote and edited all of the text, exercises and answers and created all of the graphs (using Metapost within LaTeX) for Version $0.\overline{9}$ in about eight months. (We choose to create a text in only black and white to keep printing costs to a minimum for those students who prefer a printed edition. This somewhat Spartan page layout stands in sharp relief to the explosion of colors found in most other College Algebra texts, but neither Carl nor I believe the four-color print adds anything of

value.) I used the book in three sections of College Algebra at Lorain County Community College in the Fall of 2009 and Carl's colleague, Dr. Bill Previts, taught a section of College Algebra at Lakeland with the book that semester as well. Students had the option of downloading the book as a .pdf file from our website www.stitz-zeager.com[1] or buying a low-cost printed version from our colleges' respective bookstores. (By giving this book away for free electronically, we end the cycle of new editions appearing every 18 months to curtail the used book market.) During Thanksgiving break in November 2009, many additional exercises written by Dr. Previts were added and the typographical errors found by our students and others were corrected. On December 10, 2009, Version $\sqrt{2}$ was released. The book remains free for download at our website and by using Lulu.com[2] as an on-demand printing service, our bookstores are now able to provide a printed edition for just under $19. Neither Carl nor I have, or will ever, receive any royalties from the printed editions. As a contribution back to the open-source community, all of the LaTeX files used to compile the book are available for free under a Creative Commons License on our website as well. That way, any-

[1] http://www.stitz-zeager.com

[2] http://www.lulu.com/content/paperback-book/college-algebra/7513097

one who would like to rearrange or edit the content for their classes can do so as long as it remains free.

The only disadvantage to not working for a publisher is that we don't have a paid editorial staff. What we have instead, beyond ourselves, is friends, colleagues and unknown people in the open-source community who alert us to errors they find as they read the textbook. What we gain in not having to report to a publisher so dramatically outweighs the lack of the paid staff that we have turned down every offer to publish our book. (As of the writing of this Preface, we've had three offers.) By maintaining this book by ourselves, Carl and I retain all creative control and keep the book our own. We control the organization, depth and rigor of the content which means we can resist the pressure to diminish the rigor and homogenize the content so as to appeal to a mass market. A casual glance through the Table of Contents of most of the major publishers' College Algebra books reveals nearly isomorphic content in both order and depth. Our Table of Contents shows a different approach, one that might be labeled "Functions First." To truly use The Rule of Four, that is, in order to discuss each new concept algebraically, graphically, numerically and verbally, it seems completely obvious to us that one would need to introduce functions first. (Take a moment and compare our ordering to the classic "equations first, then the Cartesian

Plane and THEN functions" approach seen in most of the major players.) We then introduce a class of functions and discuss the equations, inequalities (with a heavy emphasis on sign diagrams) and applications which involve functions in that class. The material is presented at a level that definitely prepares a student for Calculus while giving them relevant Mathematics which can be used in other classes as well. Graphing calculators are used sparingly and only as a tool to enhance the Mathematics, not to replace it. The answers to nearly all of the computational homework exercises are given in the text and we have gone to great lengths to write some very thought provoking discussion questions whose answers are not given. One will notice that our exercise sets are much shorter than the traditional sets of nearly 100 "drill and kill" questions which build skill devoid of understanding. Our experience has been that students can do about 15-20 homework exercises a night so we very carefully chose smaller sets of questions which cover all of the necessary skills and get the students thinking more deeply about the Mathematics involved.

Critics of the Open Educational Resource movement might quip that "open-source is where bad content goes to die," to which I say this: take a serious look at what we offer our students. Look through a few sections to see if what we've written is bad content in your opinion. I see

this open-source book not as something which is “free and worth every penny”, but rather, as a high quality alternative to the business as usual of the textbook industry and I hope that you agree. If you have any comments, questions or concerns please feel free to contact me at jeff@stitz-zeager.com or Carl at carl@stitz-zeager.com.

Jeff Zeager
Lorain County Community College
January 25, 2010

# Chapter 1

# Introduction to Functions

## 1.1 Functions and their Representations

## 1.2 Constant and Linear Functions

### 1.2.1 Constant Functions

Now that we have defined the concept of a function, we'll spend the rest of Chapter 1 revisiting families of curves from prior courses in Algebra by viewing them through a 'function lens'. We start with lines and refer the reader to Section **??** for a review of the basic properties of lines. The simplest lines are vertical and horizontal lines. We leave it to the reader (see Exercise 58) to think about why we eschew vertical lines in our discussion here, and begin with a functional description of horizontal lines.

Consider the horizontal lines graphed in the $xy$-plane as shown in Figure 1.2.1, Figure 1.2.2 and Figure 1.2.3. The Vertical Line Test, Theorem **??**, tells us that each describes $y$ as a function of $x$ so the question becomes how to represent these functions algebraically. The key here is to remember that the equation relating the independent variable $x$, the dependent variable $y$, and the function $f$ is given by $y = f(x)$.

In the graph on the left, $y$ always equals 3 so we have $f(x) = 3$. Procedurally, '$f(x) = 3$' says that the rule $f$ takes the input $x$, and, regardless of that input, gives the output 3. This is an example of what is called a **constant** function -

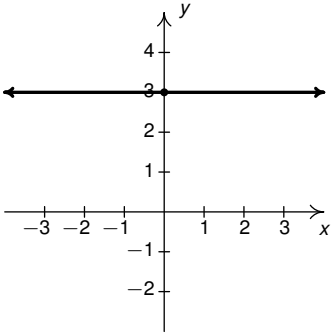



**Figure 1.2.1:** $y = 3$

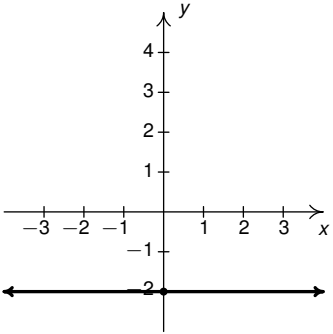



**Figure 1.2.2:** $y = -2$

a function which returns the *same* value regardless of the input. Likewise, the function represented by the graph in the middle is $f(x) = -2$, and the graph on the right (the $x$-axis) is the graph of $f(x) = 0$. In general, we have the following definition:

**Definition 1.2.1.** A **constant function** is a function of the form

$$f(x) = b$$

where $b$ is real number. The domain of a constant function is $(-\infty, \infty)$.

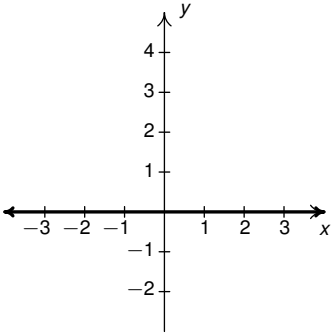



**Figure 1.2.3:** $y = 0$

Some remarks about Definition 1.2.1 are in order. First, note that we are using '$x$' as the independent variable, '$f$' as the function name, and the letter '$b$' as a **parameter**. In this context, a parameter is a fixed, but arbitrary, constant used to describe a *family* of functions. Different values of $b$ determine different constant functions. For example, $b = 3$ gives $f(x) = 3$, $b = -2$ gives $f(x) = -2$, and so on. Once $b$ is chosen, however, it does not change as the independent variable, $x$, changes.

Also note that we are using the generic defaults for function names and independent variables, namely $f$ and $x$, respectively. The functions $G(t) = \sqrt{\pi}$ and $Z(\rho) = 0$ are also fine

examples of constant functions. Recall that inherent in the definition of a function is the notion of domain, so we record (as part of the definition) that a constant function has domain $(-\infty, \infty)$. The range of a constant function is the set $\{b\}$. The value $b$ in this case is both the maximum and minimum of $f$, attained at each value in its domain.[1]

The next example showcases an application of constant functions and introduces the notion of a **piecewise-defined** function.

**Example 1.2.1.** The price of admission to see a matinee showing at a local movie theater is a function of the age of the ticket holder. If a person is aged $A$ years, the price per ticket is $p(A)$ dollars and is given by:

$$p(A) = \begin{cases} 5.75 & \text{if } 0 \leq A < 6 \text{ or } A \geq 50 \\ 7.25 & \text{if } 6 \leq A < 50 \end{cases}$$

1. Find and interpret $p(3)$, $p(6)$ and $p(62)$.
2. Explain the pricing structure verbally.
3. Graph $p$.

[1] It gets much weirder than that as we explore other more complicated functions. The key is to pay attention to the precision in the definitions of the terms involved in the discussion. Stay tuned!

**Solution.** The function $p$ described above is an example of a **piecewise-defined** function because the *rule* to determine outputs, not just the value of the output, changes depending on the inputs.

1. To find $p(3)$, we note that the value $A = 3$ satisfies the inequality $0 \leq A < 6$ so we use the rule $p(A) = 5.75$. Hence, $p(3) = 5.75$ which means a ticket for a 3 year old is \$5.75. The next age, $A = 6$, just barely satisfies the inequality $6 \leq A < 50$ so we use the rule $p(A) = 7.25$, This yields $p(6) = 7.25$ which means a ticket for a 6 year old is \$7.25. Lastly, $A = 62$ satisfies the inequality $A \geq 50$, so we are back to the rule $p(A) = 5.75$. Thus $p(62) = 5.75$ which means someone 62 years young gets in for \$5.75.

2. Now that we've had some practice interpreting function values, we can begin to verbalize what the function is really saying. In the first 'piece' of the function, the inequality $0 \leq A < 6$ describes ticket holders under the age of 6 years and the inequality $A \geq 50$ describes ticket holders fifty years old or or older. For folks in these two age demographics, $p(A) = 5.75$ so the price per ticket is \$5.75. For everyone else, that is for folks at least 6 but younger than 50, the price is \$7.25 per ticket.

3. The independent variable here is specified as $A$, so we'll label our horizontal axis that way as shown in Figure 1.2.4. The dependent variable remains unspecified so we can use the default $y$. The graph of $y = p(A)$ consists of three horizontal line pieces: the first is $y = 5.75$ for $0 \leq A < 6$, the second piece is $y = 7.25$ for $6 \leq A < 50$, and the last piece is $y = 5.75$ for $A \geq 50$.

   For the first piece, note that $A = 0$ is included in the inequality $0 \leq A < 6$ but $A = 6$ is not. For this reason, we have a point indicated at $(0, 5.75)$ but leave a hole[2] at $(6, 5.75)$. Similarly, to graph the second piece, we begin with a point at $(6, 7.25)$ and continue the horizontal line to a hole at $(50, 7.25)$. Lastly, we finish the graph with a point at $(50, 5.75)$ and continue to the right indefinitely.[3] Note the scaling on the horizontal axis compared to the vertical axis. □

One of the favorite piecewise-defined functions in mathematical circles is the **greatest integer of *x***, denoted by $\lfloor x \rfloor$. In **??** we defined the set of **integers** as $\mathbb{Z} = \{\ldots, -3, -2, -1, 0,$

[2]See our discussion about holes in graphs in Example **??** in Section **??**.

[3]The domain of $p$ is $[0, \infty)$ by definition, even though few 327 year olds are out and about these days.

[4]The use of the letter $\mathbb{Z}$ for the integers is ostensibly because the

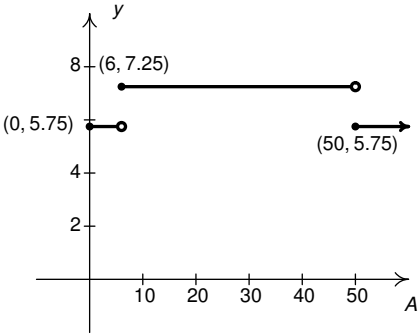



**Figure 1.2.4:** $y = p(A) = \begin{cases} 5.75 & \text{if } 0 \leq A < 6 \text{ or } A \geq 50 \\ 7.25 & \text{if } 6 \leq A < 50 \end{cases}$

The value $\lfloor x \rfloor$ is defined to be the largest integer $k$ with $k \leq x$. That is, $\lfloor x \rfloor$ is the unique integer $k$ such that $k \leq x < k + 1$. Said differently, given any real number $x$, if $x$ is an integer, then $\lfloor x \rfloor = x$. If not, then $x$ lies in an interval between two integers, $k$ and $k + 1$ and we choose $\lfloor x \rfloor = k$, the left endpoint.

**Example 1.2.2.** Let $\lfloor x \rfloor$ denote the greatest integer function.

1. Find $\lfloor 0.785 \rfloor$, $\lfloor 117 \rfloor$, $\lfloor -2.001 \rfloor$ and $\lfloor \pi + 6 \rfloor$

German word *zahlen* means 'to count'.

2. Explain how we can view $\lfloor x \rfloor$ as a piecewise-defined function and use this to graph $y = \lfloor x \rfloor$.

**Solution.**

1. To find $\lfloor 0.785 \rfloor$, we note that $0 \leq 0.785 < 1$ so $\lfloor 0.785 \rfloor = 0$. Given that 117 is an integer, we have $\lfloor 117 \rfloor = 117$. To find $\lfloor -2.001 \rfloor$, we note that $-3 \leq -2.001 < -2$, so $\lfloor -2.001 \rfloor = -3$. Finally, with $\pi \approx 3.14$, we get $\pi + 6 \approx 9.14$ and $9 \leq \pi + 6 < 10$ so $\lfloor \pi + 6 \rfloor = 9$.

2. The first step in evaluating $\lfloor x \rfloor$ is to determine the interval $[k, k + 1)$ containing $x$ so it seems reasonable that these are the intervals which produce the 'pieces'. In this case, there happen to be infinitely many pieces. The inequality '$k \leq x < k + 1$' includes the left endpoint but excludes the right endpoint, so we have points at the left endpoints of our horizontal line segments while we have holes at the right endpoints.

   A partial description of $\lfloor x \rfloor$ is given in Figure 1.2.5 and a partial graph in Figure 1.2.6. (A full description or a complete graph would require infinitely large paper!) We use the vertical dots $\vdots$ to indicate that both the rule and the graph continue indefinitely following

the established pattern.[5] □

### 1.2.2 Linear Functions

Now that we've discussed the functions which correspond to horizontal lines, $y = b$, we move to discussing the functions which can be represented by lines of the form $y = mx + b$ where $m \neq 0$. These functions are called **linear** functions and are described below.

**Definition 1.2.2.** A **linear function** is a function of the form

$$f(x) = mx + b,$$

where $m$ and $b$ are real numbers with $m \neq 0$. The domain of a linear function is $(-\infty, \infty)$.

As with Definition 1.2.1, in Definition 1.2.2, $x$ is the independent variable, $f$ is the function name, and both $m$ and $b$ are parameters. Notice that $m$ is restricted by $m \neq 0$ for if $m = 0$ then the function $f(x) = mx + b$ would reduce to the constant function $f(x) = b$. The domain of linear functions, like that of constant functions, is specified as $(-\infty, \infty)$

[5] It is always dangerous to leave the rest of the pattern to the reader. See, for instance, this paper.[6]

$$\lfloor x \rfloor = \begin{cases} \vdots & \\ -5 & \text{if } -5 \leq x < -4 \\ -4 & \text{if } -4 \leq x < -3 \\ -3 & \text{if } -3 \leq x < -2 \\ -2 & \text{if } -2 \leq x < -1 \\ -1 & \text{if } -1 \leq x < 0 \\ 0 & \text{if } \quad 0 \leq x < 1 \\ 1 & \text{if } \quad 1 \leq x < 2 \\ 2 & \text{if } \quad 2 \leq x < 3 \\ 3 & \text{if } \quad 3 \leq x < 4 \\ 4 & \text{if } \quad 4 \leq x < 5 \\ 5 & \text{if } \quad 5 \leq x < 6 \\ \vdots & \end{cases}$$

**Figure 1.2.5:** Partial description of $\lfloor x \rfloor$

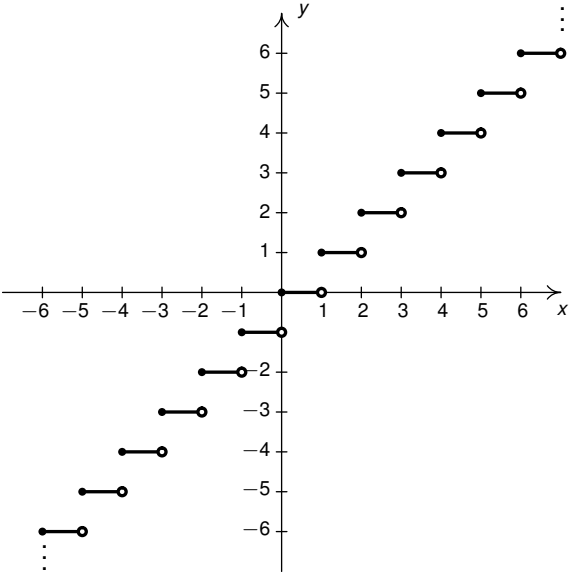



**Figure 1.2.6:** The graph of $y = \lfloor x \rfloor$

Recall[7] that the form of the line $y = mx + b$ is called the slope-intercept form of the line and the slope, $m$, and the $y$-intercept $(0, b)$, are easily determined when the line is written this way. Likewise, the form of the function in Definition 1.2.2, $f(x) = mx + b$, is often called the **slope-intercept form** of a linear function.

The graph of a linear function is the graph of the line $y = mx + b$. Lines are uniquely determined by two points, and two points of geometric interest are the axis intercepts. We've already reminded you of the $y$-intercept, $(0, b)$, which is obtained by setting $x = 0$. Similarly, to find the $x$-intercept, we set $y = 0$ and solve $mx + b = 0$ for $x$. We leave this to the reader in Exercise 38. In addition to having special graphical significance, axis intercepts quite often play important roles in applications involving both linear and nonlinear functions. For that reason, we take the time to define them here using function notation.

[7] or see Section **??**

**Definition 1.2.3.** Suppose $f$ is a function represented by the graph of $y = f(x)$.

- If 0 is in the domain of $f$ then the point $(0, f(0))$ is the **$y$-intercept** of the graph of $y = f(x)$.
  That is, $(0, f(0))$ is where the graph meets the $y$-axis.
- If 0 is in the range of $f$ then the solutions to $f(x) = 0$ are called the **zeros** of $f$. If $c$ is a zero of $f$ then the point $(c, 0)$ is an **$x$-intercept** of the graph of $y = f(x)$.
  That is, $(c, 0)$ is where the graph meets the $x$-axis.

As is customary in this text, Definition 1.2.3 uses the default independent variable $x$, function name $f$, and dependent variable $y$, so these letters will change depending on the context. Also note that the 'zeros' of a function are the solutions to $f(x) = 0$ - so they are *real numbers*. The $x$-intercepts are, on the other hand, *points* on the graph. As a quick example, consider $f(x) = x - 3$. The zeros of $f$ are found by solving $f(x) = 0$, or $x - 3 = 0$. We get one solution, $x = 3$. Therefore, $x = 3$ is the *zero* of $f$ that corresponds graphically to the *$x$-intercept* $(3, 0)$.

We now turn our attention to slope. The role of slope, or

more generally a 'rate of change', in Science and Mathematics cannot be overstated.[8] As you may recall, or quickly read about on page **??**, the slope of a line that has been graphed in the $xy$-plane is defined geometrically as follows:

$$m = \frac{\text{rise}}{\text{run}} = \frac{\Delta y}{\Delta x},$$

where the capital Greek letter '$\Delta$' denotes 'change in.'[9] In this course, it is vital that we regard the slope of a linear function as a rate of change of *function outputs* to *function inputs*. That is, given the graph of a linear function $y = f(x) = mx + b$:

$$m = \frac{\text{rise}}{\text{run}} = \frac{\Delta y}{\Delta x} = \frac{\Delta[f(x)]}{\Delta x} = \frac{\Delta\text{outputs}}{\Delta\text{inputs}}.$$

What is important to note here is that for linear functions, the rate of change $m$ is constant for all values in the domain.[10] We'll see the importance of this statement in the upcoming examples.

Geometrically, the sign of the slope has a profound impact on the graph of the line. Recall that if the slope $m > 0$, the

[8] The first half of any introductory Calculus course is about slope.

[9] More specifically, if $(x_0, y_0)$ and $(x_1, y_1)$ are two distinct points in the plane, then $\Delta x = x_1 - x_0$ and $\Delta y = y_1 - y_0$.

[10] See Exercise 57 for more details.

line rises as we read from left to right; if $m < 0$, the line falls as we read from left to right; if $m = 0$, we have a horizontal line and the graph plateaus. We define these notions more precisely for general functions in the following definition.

**Definition 1.2.4.** Let $f$ be a function defined on an interval $I$. Then $f$ is said to be:

- **increasing** on $I$ if, whenever $a < b$, then $f(a) < f(b)$. (i.e., as inputs increase, outputs **increase**.)
  **NOTE:** The graph of an increasing function **rises** as one moves from left to right.
- **decreasing** on $I$ if, whenever $a < b$, then $f(a) > f(b)$. (i.e., as inputs increase, outputs **decrease**.)
  **NOTE:** The graph of a decreasing function **falls** as one moves from left to right.
- **constant** on $I$ if $f(a) = f(b)$ for all $a$, $b$ in $I$. (i.e., outputs don't change with inputs.)
  **NOTE:** The graph of a function that is constant over an interval is a horizontal line.

Again, as with Definition 1.2.3, Definition 1.2.4 applies to any function, not just linear and constant functions. Also, note that, like Definition **??**, Definition 1.2.4 blurs the line

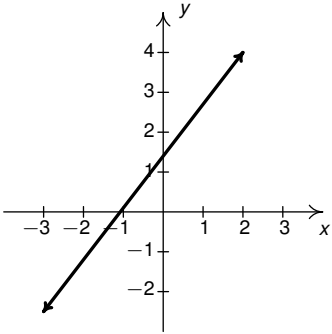



**Figure 1.2.7:** 'increasing', $m > 0$

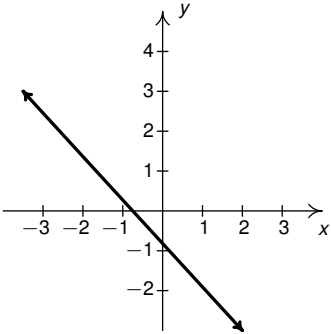



**Figure 1.2.8:** 'decreasing', $m < 0$

between the function, $f$, and its outputs, $f(x)$, because the verbiage '$f$ is increasing' is really a statement about the outputs, $f(x)$. Finally, when we ask 'where' a function is increasing, decreasing or constant, we are looking for an interval of *inputs*. We'll have more to say about this in later sections, but for now, we summarize these ideas graphically in Figure 1.2.7, Figure 1.2.8 and Figure 1.2.9.

From the graphs, we see that regardless if $m > 0$ or $m < 0$, the range of linear functions is $(-\infty, \infty)$. Therefore, linear functions have no maximum or minimum.[11]

[11] This is one of the more pedantic reasons why we distinguish be-

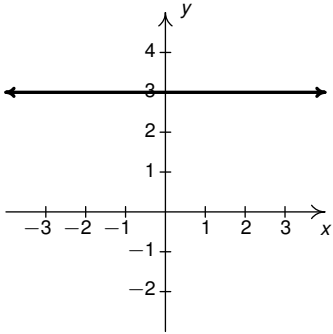



**Figure 1.2.9:** 'constant', $m = 0$

**Example 1.2.3.** The cost, in dollars, to produce $x$ PortaBoy[12] game systems for a local retailer is given by $C(x) = 80x + 150$ for $x \geq 0$.

1. Find and interpret $C(0)$ and $C(5)$ and use these to graph $y = C(x)$.
2. Explain the significance of the restriction on the domain, $x \geq 0$.
3. Interpret the slope of $y = C(x)$ geometrically and as a rate of change.

tween constant and linear functions. See the discussion concerning the range of a constant function on page 5.

[12]The similarity of this name to PortaJohn[13] is deliberate.

4. How many PortaBoys can be produced for \$15,000?

**Solution.**

1. To find $C(0)$, we substitute 0 for $x$ in the formula $C(x)$ and obtain: $C(0) = 80(0) + 150 = 150$. Given that $x$ represents the number of PortaBoys produced and $C(x)$ represents the cost to produce said PortaBoys, $C(0) = 150$ means it costs \$150 even if we don't produce any PortaBoys at all. At first, this may not seem realistic, but that \$150 is often called the **fixed** or **start-up** cost of the venture. Things like re-tooling equipment, leasing space, or any other 'up front' costs get lumped into the fixed cost. To find $C(5)$, we substitute 5 for $x$ in the formula $C(x)$: $C(5) = 80(5) + 150 = 550$. This means it costs \$550 to produce 5 PortaBoys for the local retailer. These two computations give us two points on the graph: $(0, C(0))$ and $(5, C(5))$. Along with the domain restriction $x \geq 0$, we get the graph shown in Figure 1.2.10

2. In this context, $x$ represents the number of PortaBoys produced. It makes no sense to produce a negative quantity of game systems,[14] so $x \geq 0$.

[14] Actually, it makes no sense to produce a fractional part of a game system, either, which we'll discuss later in this example.

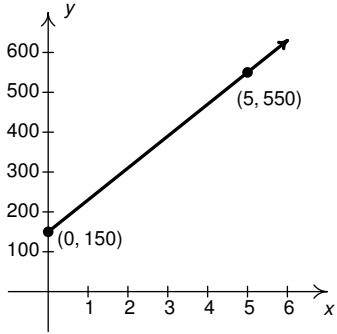



**Figure 1.2.10:** $y = C(x)$

3. The cost function $C(x) = 80x+150$ is in slope-intercept form so we recognize the slope as the coefficient of $x$, $m = 80$. With $m > 0$, the function $C$ is always increasing. This means that it costs more money to make more game systems. To interpret the slope as a rate of change, we note that the output, $C(x)$, is the cost in dollars, while the input, $x$, is the number of PortaBoys produced:

$$m = 80 = \frac{80}{1} = \frac{\Delta[C(x)]}{\Delta x} = \frac{\$80}{1 \text{ PortaBoy produced}}.$$

Hence, the cost to produce PortaBoys is increasing at a rate of \$80 per PortaBoy produced. This is often

called the **variable cost** for the venture.

4. To find how many PortaBoys can be produced for \$15,000, we solve $C(x) = 15000$, which means $80x + 150 = 15000$. This yields $x = 185.625$. We can produce only a whole number amount of PortaBoys so we are left with two options: produce 185 or 186 PortaBoys. Given that $C(185) = 14950$ and $C(186) = 15030$, we would be over budget if we produced 186 PortaBoys. Hence, we can produce 185 PortaBoys for \$15,000 (with \$50 to spare). □

A couple of remarks about Example 1.2.3 are in order. First, if $x$ represents the number of PortaBoy game systems being produced, then $x$ can really only take on whole number values. We will revisit this scenario in Section **??** where we will see how the approach presented here allows us to use more elegant techniques when analyzing the situation than a discrete data set would allow.[15]

Second, once we know that the variable cost is \$80 per PortaBoy, we can revisit a computation we did earlier in the example. We computed $C(185) = 14950$ and needed to compute $C(186)$. With 186 being just one more PortaBoy

[15] This is an example of using a 'continuous' variable to model a 'discrete' scenario. Contrast this with the discussion following Example **??** in Section **??**.

than 185, we can use the variable cost to get

$$C(186) = C(185) + 80(1) = 14950 + 80 = 15030,$$

which agrees with our earlier computation.[16] If we wanted to find $C(300)$, we could do something similar. Using $300 - 185 = 115$, we can find $C(300)$ as follows:

$$C(300) = C(185) + 80(115) = 14950 + 9200 = 24150.$$

In general, we could rewrite $C(x) = C(185) + 80(x - 115)$. This same reasoning shows that for any $x_0$ in the domain of $C$, we have $C(x) = C(x_0) + 80(x - x_0)$ - a fact we invite the reader to verify.[17]

Indeed, the computations above are at the heart of what it means to be a linear function: linear functions change at a constant rate known as the slope. To better see this algebraically, recall that given a point $(x_0, y_0)$ on a line along with the slope, $m$, the **point-slope form of the line** is: $y -$

[16] The cost to produce 'just one more item' is called the **marginal cost**. The difference between variable and marginal costs in this case are the units used: the variable cost is \$80 per Portaboy whereas the marginal cost is simply \$80.

[17] In the case $x_0 = 0$, this formula reduces to $C(x) = C(0) + 80(x - 0) = 150 + 80x = 80x + 150$. To show the formula in general, consider $C(x_0) = 80x_0 + 150 \ldots$

$y_0 = m(x - x_0)$.[18] Rewriting, we get $y = y_0 + m(x - x_0)$ and setting $y = f(x)$ and $y_0 = f(x_0)$ yields:

**Equation 1.2.1.** The **point-slope form** of a linear function is

$$f(x) = f(x_0) + m(x - x_0)$$

A few remarks are in order. First note that if the point $(x_0, f(x_0))$ is the $y$-intercept $(0, b)$, Equation 1.2.1 immediately reduces to the slope-intercept form of the line: $f(x) = f(x_0) + m(x - x_0) = b + m(x - 0) = mx + b$, so you can use Equation 1.2.1 exclusively from this point forward.[19]

Second, if we write $\Delta x = x - x_0$, then $x = x_0 + \Delta x$ so we can rewrite Equation 1.2.1 as follows:

$$\begin{array}{ccccc} f(x_0 + \Delta x) & = & f(x_0) & + & m\Delta x \\ \text{(new output)} & = & \text{(known output)} & + & \text{(change in outputs)} \end{array}$$

In other words, changing the *input* by $\Delta x$ results in changing the *output* by $m\Delta x$. This tracks since

$$m\Delta x = \frac{\Delta[f(x)]}{\Delta x}\Delta x = \Delta[f(x)] = \Delta\text{outputs}.$$

[18] See Section **??** for a review of this form.

[19] In other words, the slope intercept form of a line is just a special case of the point-slope form.

The fact that we can write $\Delta\text{outputs} = m\Delta x$ for any choice of $x_0$ is another way to see that for linear functions, the rate of change is constant. That is, the rate of change, $m$, is the same for all values $x_0$ in the domain. We'll put Equation 1.2.1 to good use in the next example.

**Example 1.2.4.** The local retailer in Example 1.2.3 is trying to mathematically model the relationship between the number of PortaBoy systems sold and the price per system. Suppose 20 systems were sold when the price was \$220 per system but when the systems went on sale for \$190 each, sales doubled.

1. Find a formula for a linear function $p$ which represents the price $p(x)$ as a function of the number of systems sold, $x$. Graph $y = p(x)$, find and interpret the intercepts, and determine a reasonable domain for $p$.
2. Interpret the slope of $p(x)$ in terms of price and game system sales.
3. If the retailer wants to sell 150 PortaBoys next week, what should the price be?
4. How many systems would sell if the price per system were set at \$150?

**Solution.**

1. We are asked to find a linear function $p(x)$ ostensibly because the retailer has only two data points and two points are all that is needed to determine a unique line. We know that 20 PortaBoys were sold when the price was 220 dollars and double that, so 40 units, were sold when the price was 190 dollars. Using the language of function notation, these statements translate to $p(20) = 220$ and $p(40) = 190$, respectively. We first find the slope

$$m = \frac{\Delta[p(x)]}{\Delta x} = \frac{190 - 220}{40 - 20} = \frac{-30}{20} = -1.5$$

and then substitute it and a pair $(x_0, p(x_0))$ into the point-slope formula. We have two choices: $x_0 = 20$ and $p(x_0) = 220$ or $x_0 = 40$ and $p(x_0) = 190$. We'll choose the former and invite the reader to use the latter - both will result in the same simplified expression. The point-slope formula yields

$$p(x) = p(x_0) + m(x - x_0) = 220 + (-1.5)(x - 40)$$

which simplifies to $p(x) = -1.5x + 250$. (To check this algebraically, we can verify that $p(20) = 220$ and $p(40) = 190$.) To find the $y$-intercept of the graph, we substitute $x = 0$ and find $p(0) = 250$. Hence our $y$-intercept is $(0, 250)$. To find the $x$-intercept, we

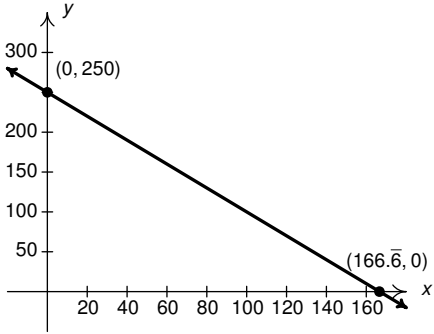



**Figure 1.2.11:** $y = -1.5x + 250$

set $p(x) = 0$. Solving $-1.5x + 250 = 0$ gives $x = 166.\overline{6}$, so our $x$-intercept is $(166.\overline{6}, 0)$.[20] The graph in Figure 1.2.11 is that of the line $y = -1.5x + 250$.

To determine a reasonable domain for $p$, we certainly require $x \geq 0$, because we can't sell a negative number of game systems.[21] Next, we require $p(x) \geq 0$, otherwise we'd be *paying* customers to 'buy' PortaBoys. Solving $-1.5x + 250 \geq 0$ results in $x \leq 166.\overline{6}$. This shouldn't be too surprising since our

[20] The exact value is $x = \frac{500}{3}$. Recall that the bar over the 6 indicates that the decimal repeats. See page **??** for details.

[21] ignoring returns, that is.

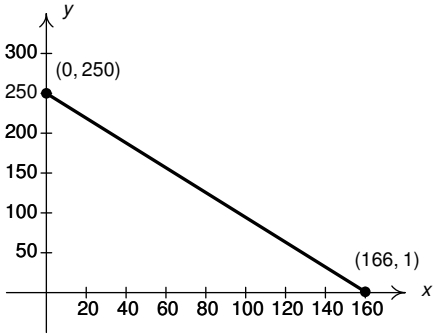



**Figure 1.2.12:** $y = p(x)$

graph passes through the $x$-axis at $(166.\overline{6}, 0)$, going from positive $y$-values (hence, positive $p(x)$ values) to negative $y$ (hence negative $p(x)$ values).[22]

Given that $x$ represents the number of PortaBoys sold, we need to choose to end the domain at either $x = 166$ or $x = 167$. We have that $p(166) = 1 > 0$ but $p(167) = -0.5 < 0$ so we settle on the domain $[0, 166]$. Our final answer is $p(x) = -1.5x + 250$ restricted to $0 \leq x \leq 166$ which is graphed in Figure 1.2.12.

[22] We'll discuss these sorts of connections in greater depth in Section **??**.

2. The slope $m = -1.5$ represents the rate of change of the price of a system with respect to sales of PortaBoys. The slope is negative so we have that the price is *decreasing* at a rate of \$1.50 per PortaBoy sold. (Said differently, you can sell one more PortaBoy for every \$1.50 drop in price.)

3. To determine the price which will move 150 PortaBoys, we find $p(150) = -1.5(150) + 250 = 25$. That is, the price would have to be \$25 per system.

4. If the price of a PortaBoy were set at \$150, we'd have $p(x) = 150$, or $-1.5x + 250 = 150$. This yields $-1.5x = -100$ or $x = 66.\overline{6}$. Again our algebraic solution lies between two whole numbers, so we find $p(66) = 151$ and $p(67) = 149.5$. If the price were set at \$150, we'd sell 66 systems, since to sell 67 systems, we'd have to drop the price just under \$150. □

The function $p$ in Example 1.2.4 is called the **price-demand** function (or, sometimes called more simply a 'demand function') because it returns the price $p(x)$ associated with a certain demand $x$ - that is, how many products will sell.[23]

[23] It may seem counter-intuitive to express price as a function of demand. Shouldn't the price determine how many systems people will buy? We will address this issue later.

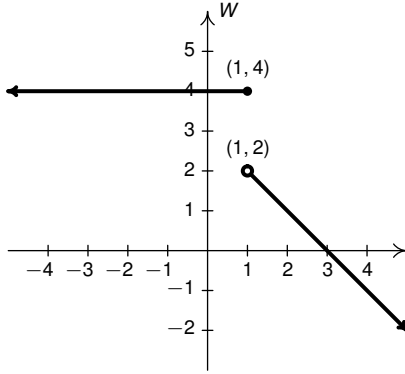



**Figure 1.2.13:** $W = L(t)$

These functions, along with cost functions like the one in Example 1.2.3, will be revisited in Example **??**.

Our next two examples focus on writing formulas for piecewise-defined functions, the second of which models a real-world situation.

**Example 1.2.5.** Find a formula for the function $L$ graphed in Figure 1.2.13.

**Solution.** From the graph of $W = L(t)$ we see that there are two distinct pieces. Taking note of the point at $(1, 4)$, we get $L(t) = 4$ for $t \leq 1$. To represent $L$ for $t > 1$, we use the point-slope form of a linear function: $L(t) = L(t_0) + m(t - t_0)$.

The only 'point' labeled with this part of the graph is the hole at $(1, 2)$ and it isn't technically part of the graph, so we will avoid using it.[24] Instead, we infer from the graph two other points: $(2, 1)$ and $(3, 0)$. We get the slope to be

$$m = \frac{\Delta W}{\Delta t} = \frac{\Delta[L(t)]}{\Delta t} = \frac{3-2}{0-1} = -1.$$

Next, we choose a point to plug into $L(t) = L(t_0) + m(t - t_0)$. We have two options: $t_0 = 2$ and $L(t_0) = 1$ or $t_0 = 3$ and $L(t_0) = 0$. Using the latter, we get $L(t) = 0 + (-1)(t - 3)$, or $L(t) = -t + 3$. Putting this together with the first part, we get:

$$L(t) = \begin{cases} 4 & \text{if } t \leq 1 \\ -t+3 & \text{if } t > 1 \end{cases}$$

Note that when $t = 1$ is substituted into the expression $-t + 3$, we get 2, so the hole at $(1, 2)$ checks.[25] □

**Example 1.2.6.** A popular Fōn-i smartphone carrier offers the following smartphone data plan: use any amount of data up to and including 4 gigabytes for \$60 per month

[24] We actually could use the point $(1, 2)$ to find the equation of the *line* containing $(1, 2)$ and, say $(3, 0)$, which is $y = -t + 3$. It's just that the graph of $L(t)$ and the line $y = -t + 3$ only agree for $t > 1$, so it would be incorrect to write $L(1) = 2$.

[25] Alternatively, for $t$ values larger than 1 but getting closer and closer to 1, $L(t) \approx 2$.

with an 'overage' charge of \$5 per gigabyte. Determine a formula that computes the cost in dollars as a function of using $g$ gigabytes of data per month. Graph your answer.

**Solution.** It is clear from context that we are to use the variable $g$ (for 'g'igabytes) as the independent variable. We are asked to compute the cost so it seems natural to name the function $C$. Hence, we are after a formula for $C(g)$. Knowing that $g$ represents the amount of data used each month, we must have $g \geq 0$. In order to get a feel for the formula for $C(g)$, we can choose some specific values for $g$ and determine the cost, $C(g)$. For example, if we use no data at all, 1 gigabyte of data, or 3.796 gigabytes of data, the cost is the same: \$60. Indeed, per the plan, for any amount of data up to and including 4 gigabytes, the cost is \$60.

Translating this to function notation means $C(0) = 60$, $C(1) = 60$, $C(3.796) = 60$, and, in general, $C(g) = 60$ for $0 \leq g \leq 4$. What happens if we use more than 4 gigabytes? Let's say we use 6 gigabytes. Per the plan, we are charged \$60 for the first 4 and then \$5 for each gigabyte over 4. Using 6 gigabytes means that we are 2 gigabytes over and our overage charge is (\$5)(2) = \$10. The total cost is the base plus the overages or \$60 + \$10 = \$70. In general, if $g > 4$, the expression $(g - 4)$ computes the amount of data

used over 4 gigabytes. Our base plus overage then comes to: $60 + 5(g - 4) = 5g + 40$. Putting this together with our previous work, we get

$$C(g) = \begin{cases} 60 & \text{if } 0 \leq g \leq 4 \\ 5g + 40 & \text{if } g > 4 \end{cases}$$

To graph $C$, we graph $y = C(g)$. For $0 \leq g \leq 4$, we have the horizontal line $y = 60$ from $(0, 60)$ to $(4, 60)$. For $g > 4$, we have the line $y = 5g + 40$. Even though the inequality $g > 4$ is strict, we nevertheless substitute $g = 4$ into the formula $y = 5g + 40$ and get $y = 60$. Normally, this would produce a hole at $(4, 60)$, but in this case, the point $(4, 60)$ is already on the graph from the first piece of the function. Essentially, the point $(4, 60)$ from $C(g) = 60$ for $0 \leq g \leq 4$ 'plugs' the hole from $C(g) = 5g + 40$ when $g > 4$.

We are graphing a line so we need to plot just one more point to determine the graph. From our work above, we know $C(6) = 70$, so we use $(6, 70)$ as our second point. Our graph is shown in Figure 1.2.14. As with the graphs shown on page **??** from Example **??**, we use '$\asymp$' to denote a break in the vertical axis in order to better display the graph.

$\square$

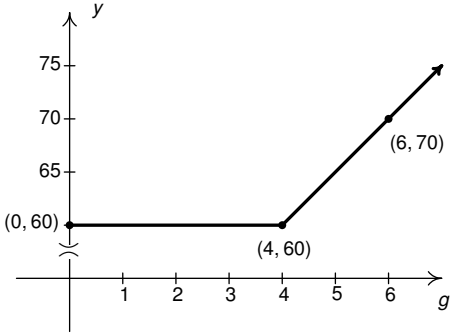



**Figure 1.2.14:** $y = C(g)$

### 1.2.3 Linear Regression

We have demonstrated in this section that constant, linear, and piecewise combinations of these two function types can be used to model a variety of phenomena inspired by real-world situations. What happens, as if often the case in real-world situations, when we are given data sets that are not precisely linear, but still have a definite linear trend? An example of this is Skippy's time and temperature data from Example **??** in Section **??** and shown in Table 1.2.1.

In that example, $t$ represented the time (number of hours after 6 a.m.) and $T$ represented the outdoor temperature in degrees Fahrenheit. The data Skippy collected along with

| $t$: hours after 6 a.m. | $T$: temperature °F |
|---|---|
| 0 | 64 |
| 2 | 67 |
| 4 | 75 |
| 6 | 80 |
| 8 | 83 |
| 10 | 83 |
| 12 | 82 |

**Table 1.2.1:** Skippy's time and temperature data

a plot of the function $T = f(t)$ are given below. Even though the data points as $t$ varies from $t = 0$ to $t = 8$ do not all lie on the same line - a fact we could prove analytically by checking slopes - there does appear to be a linear *trend* evident. The same can be said for the data as $t$ varies from $t = 8$ to $t = 12$. As we'll see, there are statistical methods which can produce linear functions that are in some sense 'closest' to all of the data, and they are represented below by the dashed lines in Figure 1.2.15.

How do we measure how 'close' a set of points is to a given line? Let's leave Skippy's data for the moment and focus on a smaller data set. Suppose we collected three data points:

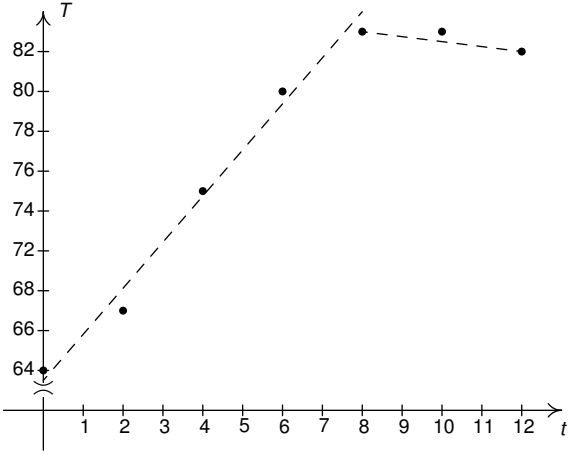



**Figure 1.2.15:** The graph of $T = f(t)$

$\{(1, 0.5), (3, 2), (4, 3)\}$. At the top of the next page (on the left) we plot these points along with the line $y = 0.5x + 0.5$. The way we measure how close the line is to these points is by computing the **total squared (vertical) error** between the data points and the line as follows. For each of our data points, we find the vertical distance between the point and the line. To accomplish this, we need to find a point on the line directly above or below each data point. In other words, we need a point on the line with the same $x$-coordinate as our data point.

For example, to find the point on the line directly above $(1, 0.5)$, we plug $x = 1$ into $y = 0.5x + 0.5$ and we get the point $(1, 1)$. Similarly, we find $(3, 2)$ is on the line already and $(4, 2.5)$ is the point on the line directly beneath $(4, 3)$. We find the total squared error $E$ by taking the sum of the squares of the differences of the $y$-coordinates of each data point and its corresponding point on the line. For the data and line in this discussion $E = (0.5 - 1)^2 + (2 - 2)^2 + (3 - 2.5)^2 = 0.5$.

Using advanced mathematical machinery,[26] it is possible to find the line which results in the lowest value of $E$. This line is called the **least squares regression line**, or sometimes the 'line of best fit'. This is shown in Figure 1.2.16. The

[26] like Calculus or Linear Algebra ...

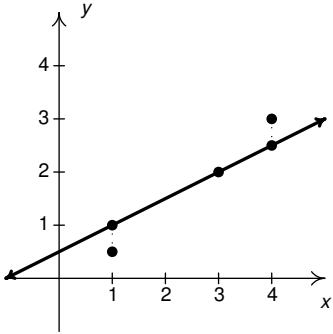



**Figure 1.2.16:** Least squares regression line

formula for the line of best fit requires notation we won't present until Chapter **??**, so we will revisit it then. Most graphing utilities have a built-in regression feature, so at this point we turn the computations over to the technology. A screenshot from desmos[27] is given in Figure 1.2.17 the right at the top of the next page.

Our graphing utility produces the model[28] $y = mx + b$ where the slope is $m \approx 0.821$ and the $y$-coordinate of the $y$-

[27] www.desmos.com

[28] We chose to use three decimal places for the approximations in this demonstration. How many you get to use in reality varies from one application to another.

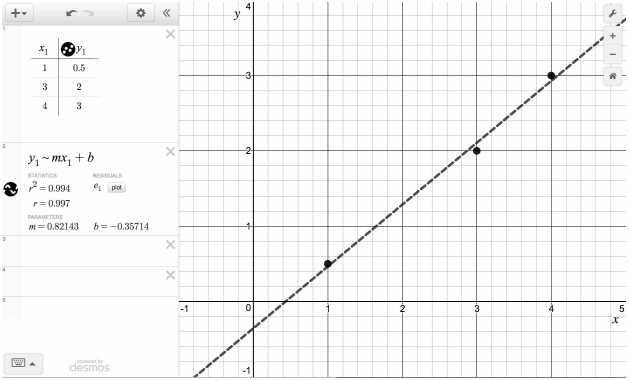



**Figure 1.2.17:** Screenshot from desmos

intercept is $b \approx -0.357$. The value $r$ is the **correlation coefficient** and is a measure of how close the data is to being on the same line. The closer $|r|$ is to 1, the better the linear fit.[29] Having $r \approx 0.997$ tells us that the points have a strong, positive correlation - that is, they are very close to being on a line with a positive slope, namely $y = 0.821x - 0.357$. Indeed, the total squared error between our data set and this line is $E \approx 0.018$. The math-

[29]The value $r^2$ is called the **coefficient of determination** and is also a measure of the goodness of fit. We refer the interested reader to a course in Statistics to explore the significance of $r$ and $r^2$.

| $t$: hours after 6 a.m. | $T$: temperature °F |
|---|---|
| 0 | 64 |
| 2 | 67 |
| 4 | 75 |
| 6 | 80 |
| 8 | 83 |

**Table 1.2.2:** Dataset 1

ematics tells us that this is the smallest we can get $E$ by modifying the parameters $m$ and $b$, even though none of the data points actually lie on the line.

Now that we have this new mathematical machinery, let's revisit Skippy's time and temperature data.

**Example 1.2.7.**

1. Use a graphing utility to find best fit linear models for each of the data sets shown in Table 1.2.2 and Table 1.2.3. Comment on the fit and interpret the slope of each.

2. Use your models to predict the temperature at 7 a.m. and 3 p.m., rounded to one decimal place.

**Solution.**

| $t$: hours after 6 a.m. | $T$: temperature °F |
|---|---|
| 8 | 83 |
| 10 | 83 |
| 12 | 82 |

**Table 1.2.3:** Dataset 2

1. For our first set of data, we get the line $T = F(t) = 2.55t + 63.6$. The value $r = 0.987$ tells us that it is a fairly good fit and we see this graphically, too.[30] Thus we can be confident in using this model to predict the temperature during between the hours of 6 a.m. and 2 p.m. with reasonable accuracy.

   To interpret the slope, we recognize $t$ as the independent variable (input) and $T$ as the dependent variable (output), so the slope $m = \frac{\Delta T}{\Delta t}$ is the rate of change of temperature with respect to time. In this case, $m = 2.55$ means that the temperature is increasing (getting warmer) at a rate of 2.55°F per hour. A screenshot from Desmos is given in Figure 1.2.18.

   For the second set of data, we get $T = G(t) = -0.25t+$

[30] We use $F$ as the name of the function here to distinguish it from $f$ - the function determined solely by the given set of data.

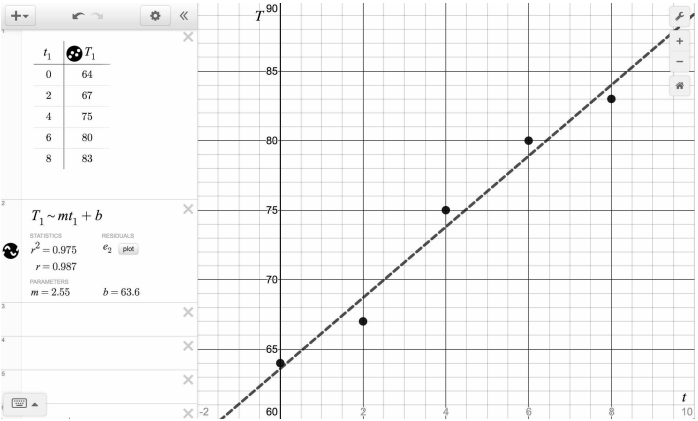



**Figure 1.2.18:** Screenshot from desmos

85.167 and we have $r = -0.866$. Here, the negative sign on $r$ indicates a negative correlation which means our line has a negative slope.[31] While the fit looks OK, it certainly isn't as strong as with the first data set, so using this model to predict the temperature between 2 p.m. and 6 p.m. (let alone beyond) is a bit risky.

The slope in this case is $m = -0.25$ which corresponds to the temperature decreasing (getting cooler) at a rate of 0.25°F per hour. That's what a negative correlation means - an increase in input (more time passes) yields a decrease in output (cooler temperatures). As shown in screenshot from Desmos, given in Figure 1.2.19.

2. The time 7 a.m. corresponds to $t = 1$. This falls between $t = 0$ and $t = 8$ so we use our first model. Substituting $t = 1$ gives $T = F(1) = 2.55(1)+63.6 \approx 66.2$. Therefore, the model predicts the temperature to be 66.2°F at 7 a.m.. Likewise, 3 p.m. corresponds to $t = 9$. This is greater than 8, so we use the second model: $T = G(9) = -0.25(9) + 85.167 \approx 82.9$. The model predicts the temperature at 3 p.m. to be

[31] We use $G$ as the function name here to distinguish it from the given function $f$ and the regression for the first data set, $F$.

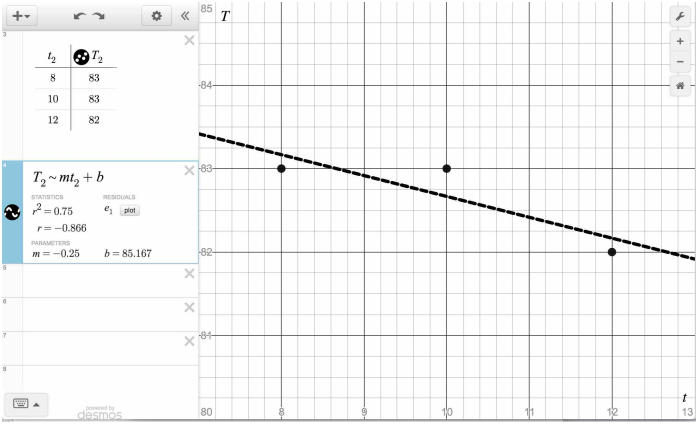



**Figure 1.2.19:** Screenshot from desmos

82.9°F. Based on the goodness of fit of each model, we have more confidence in the former prediction than in the latter. □

Examples 1.2.3, 1.2.4 and 1.2.7 (among others) represent three different levels of mathematical modeling. In Example 1.2.3, the mathematical model (the cost function) was provided and our task was to use the model to *interpret* the mathematics in that context. In Example 1.2.4, we were given a minimal amount of information, namely, two data points, and then asked to *construct* a model which fit those data exactly. Lastly, in Example 1.2.7, we were given several data points and we used statistical methods to construct a *best fit* model to the data.

The validity of the models rests on the validity of the underlying assumptions used to create the models. For instance, is there any reason to assume a price-demand function would be linear? Is it reasonable to assume that the temperature changes at a constant rate? These are questions for economists and scientists. Mathematicians often take on a role of equal parts translator and prophet: they codify ideas into formulas and then use them to make predictions about yet-to-be observed phenomena.

### 1.2.4 The Average Rate of Change of a Function

As mentioned earlier in the section, the concepts of slope and the more general rates of change are important concepts not just in Mathematics, but also in other fields. Many important phenomena are modeled using non-linear functions, and while the rates of change of these functions are not constant, we can sample the function at two points and compute what is known as an **average rate of change** between them to give some sense as to the function's behavior over that interval.[32]

**Definition 1.2.5.** Let $f$ be a function defined on the interval $[a, b]$. The **average rate of change** of $f$ over $[a, b]$ is defined as:

$$\frac{\Delta[f(x)]}{\Delta x} = \frac{f(b) - f(a)}{b - a}$$

Geometrically, the average rate of change is the slope of the line[a] containing $(a, f(a))$ and $(b, f(b))$.

[a] This line is called a *secant* line.

[32] We are basically pretending that the function is linear on a short interval to see what we can say about its behavior.

As with Definitions **??** and 1.2.4, the wording in Definition 1.2.5, while referring to the function $f$, is really making a statement about its outputs $f(x)$.

If $f$ is increasing over $[a, b]$, then the average rate of change will be positive. Likewise, if $f$ is decreasing or constant, the average rate of change will be negative or 0, respectively. (Think about this for a moment.) However, as the next example demonstrates, the converses of these statements aren't always true.[33]

**Example 1.2.8.** The formula $s(t) = -5t^2 + 100t$ for $0 \leq t \leq 20$ gives the height, $s(t)$, measured in feet, of a model rocket above the Moon's surface as a function of the time $t$, in seconds after lift-off.

1. Find $s(0)$, $s(5)$, $s(10)$, $s(15)$ and $s(20)$ and use these along with a graphing utility to graph $y = s(t)$.
2. State the range of $s$ and interpret the extrema, if any exist.
3. Find and interpret the $t$- and $y$-intercepts.
4. Find and interpret the interval(s) over which $s$ is increasing, decreasing or constant.

[33] For example, the average rate of change over an interval could be positive yet the function could decrease over part of that interval and then increase on a different part.

5. Find and interpret the average rate of change of $s$ over the intervals $[0, 5]$, $[5, 10]$, $[10, 20]$ and $[5, 15]$.

**Solution.**

1. To find $s(0)$, we substitute $t = 0$ into the formula for $s(t)$: $s(0) = -5(0)^2 + 100(0) = 0$. Similarly, $s(5) = -5(5)^2 + 100(5) = -5(25) + 500 = -125 + 500 = 375$. Continuing, we obtain: $s(10) = 500$, $s(15) = 375$ and $s(20) = 0$. Using these, we construct a table of values and with the help of a graphing utility we obtain the graph shown in Figure 1.2.20:

2. Projecting the graph to the $y$-axis, we see that the range of $s$ is $[0, 500]$ so the minimum of $s$ is 0 and the maximum is 500. This means that the rocket at some point is on the surface of the Moon and reaches its highest altitude of 500 feet above the lunar surface.

3. The first intercept we see is $(0, 0)$ which is both a $t$- and a $y$-intercept. Given that $t$ represents the time after lift-off and $y = s(t)$ represents the height above the Moon's surface, the point $(0, 0)$ means that the model rocket was launched ($t = 0$) from the Moon's surface ($s(t) = 0$). The remaining intercept, $(20, 0)$, is another $t$-intercept. This means that 20 seconds after lift-off ($t = 20$), the model rocket returns to the

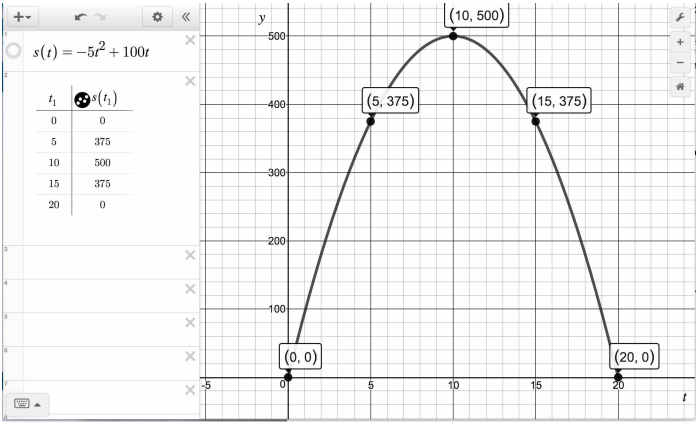



**Figure 1.2.20:** Screenshot from desmos

Moon's surface ($s(t) = 0$). Said differently, 20 seconds is the 'time of flight' of the model rocket.

4. Referring to Definition 1.2.4, $s$ increases over the interval $[0, 10]$, since for those values of $t$, as we read from left to right, the graph of the function is rising meaning the $y$ values (hence $s(t)$ values) are getting larger. Thus the model rocket is heading upwards for the first 10 seconds of its flight. We find that $s$ decreases over the interval $[10, 20]$, indicating once it has reached its highest altitude of 500 feet 10 seconds into the flight, the rocket begins to fall back to the surface of the Moon, landing 20 seconds after lift-off.

5. To find the average rate of change of $s$ over the interval $[0, 5]$ we compute

$$\frac{\Delta[s(t)]}{\Delta t} = \frac{s(5) - s(0)}{5 - 0} = \frac{375 \text{ feet}}{5 \text{ seconds}} = 75 \text{ feet per second.}$$

In other words, the height is *increasing* at an *average rate* of 75 feet per second during the first 5 seconds of flight. The rate here is called the **average velocity** of the rocket over this interval. Velocity differs from speed in that velocity comes with a direction. In this case, a positive velocity indicates that the rocket is

traveling *upwards*, since when $s$ is increasing, the model rocket is climbing higher.

Similarly, the average rate of change of $s$ over the interval $[5, 10]$ works out to be 25. This means that the average velocity over the next 5 seconds of the flight has slowed to 25 feet per second. The model rocket is still, on average, traveling upwards, albeit more slowly than before.

Over the interval $[10, 20]$, the average rate of change of $s$ works out to be $-50$. This means that, on average, the rocket is *falling* at a rate of 50 feet per second. The rocket has managed to fall from its highest point 500 feet above the surface of the Moon back to the Moon's surface in 10 seconds so this makes sense. Last, but not least, the average rate of change of $s$ over $[5, 15]$ turns out to be 0. This means that the model is the same height above the ground after 5 seconds (375 feet) as it is after 15 seconds.

Geometrically, the average rate of change of a function over an interval can be interpreted as the slope of a secant line. Figure 1.2.21 shows a dotted line containing $(0, 0)$ and $(5, 375)$ (which has slope 75) along with a dotted line containing the points $(5, 375)$ and $(10, 500)$ (which has slope 25). Visually, the lines

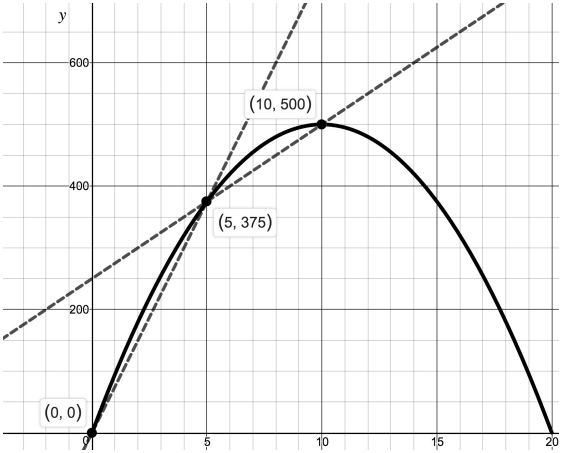



**Figure 1.2.21:** Rate of increase of s

help demonstrate that, while $s$ is increasing over $[0, 10]$, the rate of increase is slowing down as $t$ nears 10.

The graph in Figure 1.2.22 depicts a dotted line through $(10, 500)$ and $(20, 0)$ indicating a net decrease over that interval. We also have a horizontal line (0 slope) containing the points $(5, 375)$ and $(15, 375)$, which shows no net change between those two points, despite the fact that the rocket rose to its maximum height then began its descent during the interval $[5, 15]$.

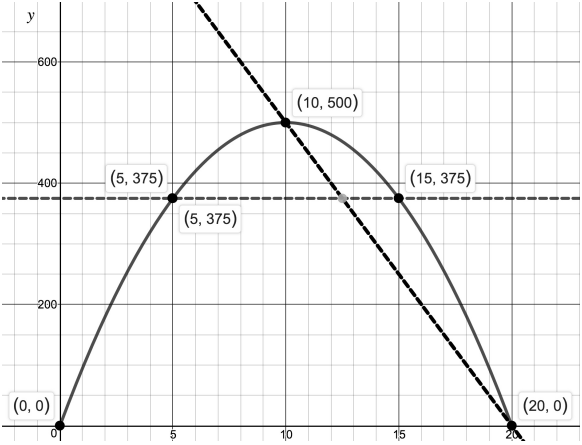



**Figure 1.2.22:** Rate of decrease of s

□

An important lesson from the last example is that average rates of change give us a snapshot of what is happening *at the endpoints* of an interval, but not necessarily what happens *over the course* of the interval. Calculus gives us tools to compute slopes *at* points which correspond to *instantaneous* rates of changes. While we don't quite have the machinery to properly express these ideas, we can hint at them in the Exercises. Speaking of exercises . . .

### 1.2.5 Exercises

In Exercises 1 - 6, graph the function. Find the slope and axis intercepts, if any.

1. $f(x) = 2x - 1$
2. $g(t) = 3 - t$
3. $F(w) = 3$
4. $G(s) = 0$
5. $h(t) = \frac{2}{3}t + \frac{1}{3}$
6. $j(w) = \dfrac{1 - w}{2}$

In Exercises 7 - 10, graph the function. Find the domain, range, and axis intercepts, if any.

7. $f(x) = \begin{cases} 4 - x & \text{if} \quad x \leq 3 \\ 2 & \text{if} \quad x > 3 \end{cases}$

8. $g(x) = \begin{cases} 2 - x & \text{if} \quad x < 2 \\ x - 2 & \text{if} \quad x \geq 2 \end{cases}$

9. $F(t) = \begin{cases} -2t - 4 & \text{if} \quad t < 0 \\ 3t & \text{if} \quad t \geq 0 \end{cases}$

10. $G(t) = \begin{cases} -3 & \text{if} \quad t < 0 \\ 2t - 3 & \text{if} \quad 0 < t < 3 \\ 3 & \text{if} \quad t > 3 \end{cases}$

11. The **unit step function** is defined as $U(t) = \begin{cases} 0 & \text{if } t < 0, \\ 1 & \text{if } t \geq 0. \end{cases}$

    (a) Graph $y = U(t)$.

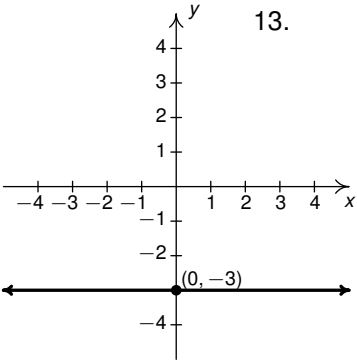
(b) State the domain and range of $U$.

(c) List the interval(s) over which $U$ is increasing, decreasing, and/or constant.

(d) Write $U(t-2)$ as a piecewise defined function and graph.

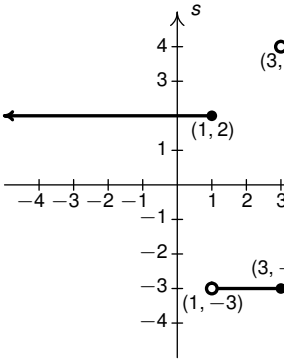
In Exercises 12. - 15., find a formula for the function.

12.



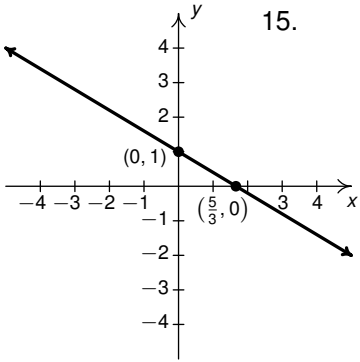

$y = f(x)$

13.



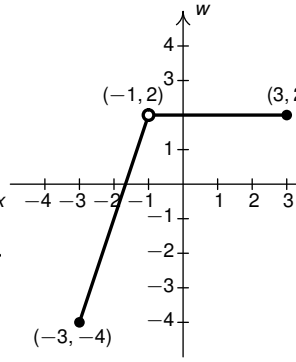

$s = F(t)$

14.




$y = L(x)$

15.




$w = g(v)$

16. For $n$ copies of the book *Me and my Sasquatch*, a print on-demand company charges $C(n)$ dollars, where $C(n)$ is determined by the formula

$$C(n) = \begin{cases} 15n & \text{if} \quad 1 \leq n \leq 25 \\ 13.50n & \text{if} \quad 25 < n \leq 50 \\ 12n & \text{if} \quad n > 50 \end{cases}$$

(a) Find and interpret $C(20)$.

(b) How much does it cost to order 50 copies of the book? What about 51 copies?

(c) Your answer to 16b should get you thinking. Suppose a bookstore estimates it will sell 50 copies of the book. How many books can, in fact, be

ordered for the same price as those 50 copies? (Round your answer to a whole number of books.)

17. An on-line comic book retailer charges shipping costs according to the following formula

$$S(n) = \begin{cases} 1.5n + 2.5 & \text{if} \quad 1 \leq n \leq 14 \\ 0 & \text{if} \quad n \geq 15 \end{cases}$$

where $n$ is the number of comic books purchased and $S(n)$ is the shipping cost in dollars.

(a) What is the cost to ship 10 comic books?

(b) What is the significance of the formula $S(n) = 0$ for $n \geq 15$?

18. The cost in dollars $C(m)$ to talk $m$ minutes a month on a mobile phone plan is modeled by

$$C(m) = \begin{cases} 25 & \text{if} \quad 0 \leq m \leq 1000 \\ 25 + 0.1(m - 1000) & \text{if} \quad m > 1000 \end{cases}$$

(a) How much does it cost to talk 750 minutes per month with this plan?

(b) How much does it cost to talk 20 hours a month with this plan?

(c) Explain the terms of the plan verbally.

19. Jeff can walk comfortably at 3 miles per hour. Find an expression for a linear function $d(t)$ that represents the total distance Jeff can walk in $t$ hours, assuming he doesn't take any breaks.

20. Carl can stuff 6 envelopes per *minute*. Find an expression for a linear function $E(t)$ that represents the total number of envelopes Carl can stuff after $t$ *hours*, assuming he doesn't take any breaks.

21. A landscaping company charges \$45 per cubic yard of mulch plus a delivery charge of \$20. Find an expression for a linear function $C(x)$ which computes the total cost in dollars to deliver $x$ cubic yards of mulch.

22. A plumber charges \$50 for a service call plus \$80 per hour. If she spends no longer than 8 hours a day at any one site, find an expression for a linear function $C(t)$ that computes her total daily charges in dollars as a function of the amount of time spent in hours, $t$ at any one given location.

23. A salesperson is paid \$200 per week plus 5% commission on her weekly sales of $x$ dollars. Find an expression for a linear function $W(x)$ which computes

her total weekly pay in dollars as a function of $x$. What must her weekly sales be in order for her to earn \$475.00 for the week?

24. An on-demand publisher charges \$22.50 to print a 600 page book and \$15.50 to print a 400 page book. Find an expression for a linear function which models the cost of a book in dollars $C(p)$ as a function of the number of pages $p$. Find and interpret both the slope of the linear function and $C(0)$.

25. The Topology Taxi Company charges \$2.50 for the first fifth of a mile and \$0.45 for each additional fifth of a mile. Find an expression for a linear function which models the taxi fare $F(m)$ as a function of the number of miles driven, $m$. Find and interpret both the slope of the linear function and $F(0)$.

26. Water freezes at $0^\circ$ Celsius and $32^\circ$ Fahrenheit and it boils at $100^\circ$C and $212^\circ$F.

    (a) Find an expression for a linear function $F(T)$ that computes temperature in the Fahrenheit scale as a function of the temperature $T$ given in degrees Celsius. Use this function to convert $20^\circ$C into Fahrenheit.

    (b) Find an expression for a linear function $C(T)$

that computes temperature in the Celsius scale as a function of the temperature $T$ given in degrees Fahrenheit. Use this function to convert 110°F into Celsius.

(c) Is there a temperature $T$ such that $F(T) = C(T)$?

27. Legend has it that a bull Sasquatch in rut will howl approximately 9 times per hour when it is $40^\circ F$ outside and only 5 times per hour if it's $70^\circ F$. Assuming that the number of howls per hour, $N$, can be represented by a linear function of temperature Fahrenheit, find the number of howls per hour he'll make when it's only $20^\circ F$ outside. What troubles do you encounter when trying to determine a reasonable applied domain?

28. Economic forces have changed the cost function for PortaBoys to $C(x) = 105x + 175$. Rework Example 1.2.3 with this new cost function.

29. In response to the economic forces in Exercise 28 above, the local retailer sets the selling price of a PortaBoy at \$250. Remarkably, 30 units were sold each week. When the systems went on sale for \$220, 40 units per week were sold. Rework Example 1.2.4 with this new data.

30. A local pizza store offers medium two-topping pizzas delivered for \$6.00 per pizza plus a \$1.50 delivery charge per order. On weekends, the store runs a 'game day' special: if six or more medium two-topping pizzas are ordered, they are \$5.50 each with no delivery charge. Write a piecewise-defined linear function which calculates the cost in dollars $C(p)$ of $p$ medium two-topping pizzas delivered during a weekend.

31. A restaurant offers a buffet which costs \$15 per person. For parties of 10 or more people, a group discount applies, and the cost is \$12.50 per person. Write a piecewise-defined linear function which calculates the total bill $T(n)$ of a party of $n$ people who all choose the buffet.

32. A mobile plan charges a base monthly rate of \$10 for the first 500 minutes of air time plus a charge of 15¢ for each additional minute. Write a piecewise-defined linear function which calculates the monthly cost in dollars $C(m)$ for using $m$ minutes of air time.

    **HINT:** You may wish to refer to number 18 for inspiration.

33. The local pet shop charges 12¢ per cricket up to 100 crickets, and 10¢ per cricket thereafter. Write

a piecewise-defined linear function which calculates the price in dollars $P(c)$ of purchasing $c$ crickets.

34. The cross-section of a swimming pool is shown in Figure 1.2.23. Write a piecewise-defined linear function which describes the depth of the pool, $D$ (in feet) as a function of:

    (a) the distance (in feet) from the edge of the shallow end of the pool, $d$.

    (b) the distance (in feet) from the edge of the deep end of the pool, $s$.

    (c) Graph each of the functions in (a) and (b). Discuss with your classmates how to transform one into the other and how they relate to the diagram of the pool.

35. The function defined by $I(x) = x$ is called the Identity Function. Thinking from a procedural perspective, explain a possible origin of this name.

36. Why must the graph of a function $y = f(x)$ have at most one $y$-intercept?

    **HINT:** Consider what would happen graphically if there were more than one . . .

37. Why is a discussion of vertical lines omitted when discussing functions?

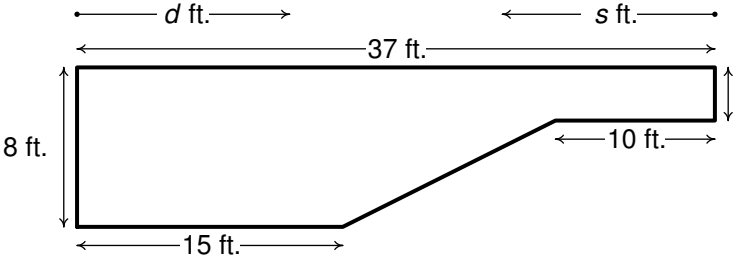



**Figure 1.2.23:** Cross-secion of swimming pool

38. Find a formula for the $x$-intercept of the graph of $f(x) = mx + b$. Assume $m \neq 0$.

39. Suppose $(c, 0)$ is the $x$-intercept of a linear function $f$. Use the point-slope form of a liner function, Equation 1.2.1 to show $f(x) = m(x - c)$. This is the 'slope $x$-intercept' form of the linear function.

40. Prove that for all linear functions $L$ with with slope 3, $L(120) = L(100) + 60$.

41. Find the slopes between the following points from the data set given in Example 1.2.7 and compare them with the slope of the corresponding regression line:

    (a) $(0, 64), (4, 75)$ (b) $(4, 75), (8, 83)$
    (c) $(8, 83), (10, 83)$

(d) $(10, 83), (12, 82)$

42. According to this website[34][35], the census data for Lake County, Ohio is:

| Year | 1970 | 1980 | 1990 | 2000 |
|---|---|---|---|---|
| Population | 197200 | 212801 | 215499 | 227511 |

(a) Find the least squares regression line for these data and comment on the goodness of fit.[37] Interpret the slope of the line of best fit.

(b) Use the regression line to predict the population of Lake County in 2010. (The recorded figure from the 2010 census is 230,041)

(c) Use the regression line to predict when the population of Lake County will reach 250,000.

43. According to this website[38][39], the census data for Lorain County, Ohio is:

[34] http://www.ohiobiz.com/census/Lake.pdf

[35] http://www.ohiobiz.com/census/Lake.pdf[36]

[37] We'll develop more sophisticated models for the growth of populations in Chapter **??**. For the moment, we use a theorem from Calculus to approximate those functions with lines.

[38] http://www.ohiobiz.com/census/Lorain.pdf

[39] http://www.ohiobiz.com/census/Lorain.pdf[40]

| Year | 1970 | 1980 | 1990 | 2000 |
|---|---|---|---|---|
| Population | 256843 | 274909 | 271126 | 284664 |

(a) Find the least squares regression line for these data and comment on the goodness of fit. Interpret the slope of the line of best fit.

(b) Use the regression line to predict the population of Lorain County in 2010. (The recorded figure from the 2010 census is 301,356)

(c) Use the regression line to predict when the population of Lake County will reach 325,000.

44. Table 1.2.4 contains a portion of the fuel consumption information for a 2002 Toyota Echo that Jeffrey used to own. The first row is the cumulative number of gallons of gasoline that I had used and the second row is the odometer reading when I refilled the gas tank. So, for example, the fourth entry is the point (28.25, 1051) which says that I had used a total of 28.25 gallons of gasoline when the odometer read 1051 miles.

    Find the least squares line for this data. Is it a good fit? What does the slope of the line represent? Do you and your classmates believe this model would

| Gasoline Used (Gallons) | Odometer (Miles) |
|---:|---:|
| 0 | 41 |
| 9.26 | 356 |
| 19.03 | 731 |
| 28.25 | 1051 |
| 36.45 | 1347 |
| 44.64 | 1631 |
| 53.57 | 1966 |
| 62.62 | 2310 |
| 71.93 | 2670 |
| 81.69 | 3030 |
| 90.43 | 3371 |

**Table 1.2.4:** Fuel consumption of 2002 Toyota Echo

have held for ten years had I not crashed the car on the Turnpike a few years ago?

45. Using the energy production data given below

| Year | 1950 | 1960 | 1970 | 1980 | 1990 | 20 |
|---|---|---|---|---|---|---|
| Production (in Quads) | 35.6 | 42.8 | 63.5 | 67.2 | 70.7 | 71 |

(a) Plot the data using a graphing utility and explain why it does not appear to be linear.

(b) Discuss with your classmates why ignoring the first two data points may be justified from a historical perspective.

(c) Find the least squares regression line for the last four data points and comment on the goodness of fit. Interpret the slope of the line of best fit.

(d) Use the regression line to predict the annual US energy production in the year 2010.

(e) Use the regression line to predict when the annual US energy production will reach 100 Quads.

In Exercises 46 - 51, compute the average rate of change of the function over the specified interval.

46. $f(x) = x^3$, $[-1, 2]$

47. $g(x) = \dfrac{1}{x}$, $[1, 5]$

48. $f(t) = \sqrt{t}$, $[0, 16]$
49. $g(t) = x^2$, $[-3, 3]$
50. $F(s) = \dfrac{s+4}{s-3}$, $[5, 7]$
51. $G(s) = 3s^2 + 2s - 7$, $[-4, 2]$
52. The height of an object dropped from the roof of a building is modeled by: $h(t) = -16t^2 + 64$, for $0 \leq t \leq 2$. Here, $h(t)$ is the height of the object off the ground in feet $t$ seconds after the object is dropped. Find and interpret the average rate of change of $h$ over the interval $[0, 2]$.
53. Using data from Bureau of Transportation Statistics[41], the average fuel economy $F(t)$ in miles per gallon for passenger cars in the US can be modeled by $F(t) = -0.0076t^2 + 0.45t + 16$, $0 \leq t \leq 28$, where $t$ is the number of years since 1980. Find and interpret the average rate of change of $F$ over the interval $[0, 28]$.
54. The temperature $T(t)$ in degrees Fahrenheit $t$ hours after 6 AM is given by:

[41] http://www.bts.gov/publications/national_transportation_statistics/html/table_04_23.html

$$T(t) = -\frac{1}{2}t^2 + 8t + 32, \quad 0 \leq t \leq 12$$

(a) Find and interpret $T(4)$, $T(8)$ and $T(12)$.

(b) Find and interpret the average rate of change of $T$ over the interval $[4, 8]$.

(c) Find and interpret the average rate of change of $T$ from $t = 8$ to $t = 12$.

(d) Find and interpret the average rate of temperature change between 10 AM and 6 PM.

55. Suppose $C(x) = x^2 - 10x + 27$ represents the costs, in *hundreds*, to produce $x$ *thousand* pens. Find and interpret the average rate of change as production is increased from making 3000 to 5000 pens.

56. Recall from Example 1.2.8 The formula $s(t) = -5t^2 + 100t$ for $0 \leq t \leq 20$ gives the height, $s(t)$, measured in feet, of a model rocket above the Moon's surface as a function of the time after lift-off, $t$, in seconds.

    (a) Find and interpret the average rate of change of $s$ over the following intervals:

    i. $[14.9, 15]$ ii. $[15, 15.1]$
    iii. $[14.99, 15]$ iv. $[15, 15.01]$

(b) What value does the average rate of change appear to be approaching as the interval shrinks closer to the value $t = 15$?

(c) Find the equation of the line containing $(15, 375)$ with slope $m = -50$ and graph it along with $s$ on the same set of axes using a graphing utility. What happens as you zoom in near $(15, 375)$?

57. Show the average rate of change of a function of the form $f(x) = mx + b$ over *any* interval is $m$.

58. Why doesn't the graph of the vertical line $x = b$ in the $xy$-plane represent $y$ as a function of $x$?

59. With help from a graphing utility, graph the following pairs of functions on the same set of axes:[42]

    - $f(x) = 2 - x$ and $g(x) = \lfloor 2 - x \rfloor$
    - $f(x) = x^2 - 4$ and $g(x) = \lfloor x^2 - 4 \rfloor$
    - $f(x) = x^3$ and $g(x) = \lfloor x^3 \rfloor$
    - $f(x) = \sqrt{x} - 4$ and $g(x) = \lfloor \sqrt{x} - 4 \rfloor$

    Choose more functions $f(x)$ and graph $y = f(x)$ alongside $y = \lfloor f(x) \rfloor$ until you can explain how, in general, one would obtain the graph of $y = \lfloor f(x) \rfloor$ given the graph of $y = f(x)$.

[42] See Example 1.2.2 for the definition of $\lfloor x \rfloor$.

60. The Lagrange Interpolate[43] function $L$ for two points $(x_0, y_0)$ and $(x_1, y_1)$ where $x_0 \neq x_1$ is given by:

$$L(x) = y_0 \frac{x - x_1}{x_0 - x_1} + y_1 \frac{x - x_0}{x_1 - x_0}$$

(a) For each of the following pairs of points, find $L(x)$ using the formula above and verify each of the points lies on the graph of $y = L(x)$.

i. $(-1, 3), (2, 3)$ ii. $(-3, -2), (5, -2)$
iii. $(-3, -2), (0, 1)$
iv. $(-1, 5), (2, -1)$

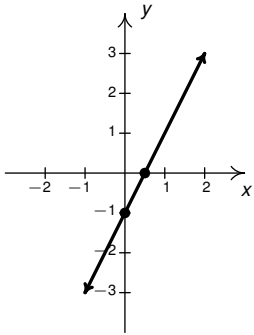
(b) Verify that, in general, $L(x_0) = y_0$ and $L(x_1) = y_1$.

(c) Show the point-slope form of a linear function, Equation 1.2.1 is equivalent to the formula given for $L(x)$ after making the identifications: $f(x_0) = y_0$ and $m = \dfrac{y_1 - y_0}{x_1 - x_0}$.

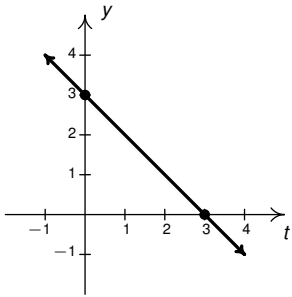
[43] https://en.wikipedia.org/wiki/Lagrange_polynomial

### 1.2.6 Answers

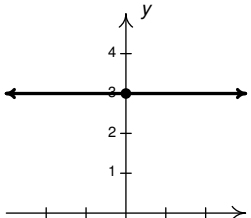
1. $f(x) = 2x - 1$
   slope: $m = 2$
   $y$-intercept: $(0, -1)$
   $x$-intercept: $\left(\frac{1}{2}, 0\right)$



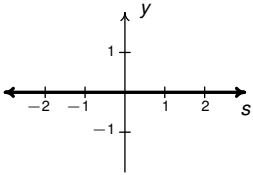

2. $g(t) = 3 - t$
   slope: $m = -1$
   $y$-intercept: $(0, 3)$
   $t$-intercept: $(3, 0)$



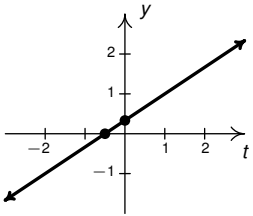

3. $F(w) = 3$
   slope: $m = 0$
   $y$-intercept: $(0, 3)$
   $w$-intercept: none



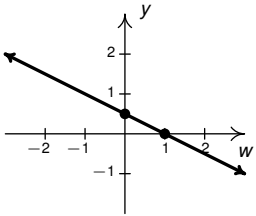
4. $G(s) = 0$
   slope: $m = 0$
   $y$-intercept: $(0, 0)$
   $s$-intercept:
   $\{(s, 0) \mid s \text{ is a real number}\}$



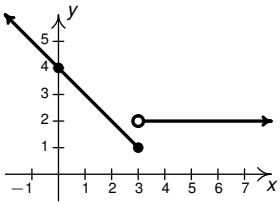

5. $h(t) = \frac{2}{3}x + \frac{1}{3}$
   slope: $m = \frac{2}{3}$
   $y$-intercept: $\left(0, \frac{1}{3}\right)$
   $t$-intercept: $\left(-\frac{1}{2}, 0\right)$



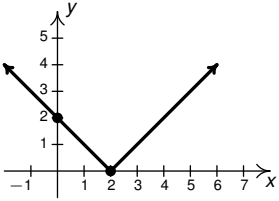

6. $j(w) = \dfrac{1 - w}{2}$
   slope: $m = -\frac{1}{2}$
   $y$-intercept: $\left(0, \frac{1}{2}\right)$
   $w$-intercept: $(1, 0)$



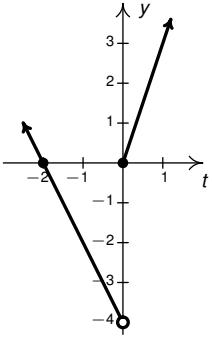

7. domain: $(-\infty, \infty)$
   range: $[1, \infty)$
   $y$-intercept: $(0, 4)$
   $x$-intercept: none



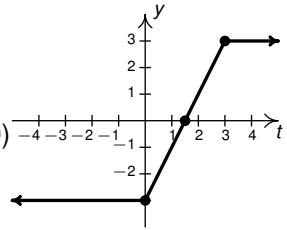
8. domain: $(-\infty, \infty)$
   range: $[0, \infty)$
   $y$-intercept: $(0, 2)$
   $x$-intercept: $(2, 0)$



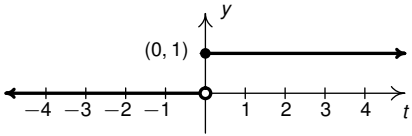

9. domain: $(-\infty, \infty)$
   range: $(-4, \infty)$
   $y$-intercept: $(0, 0)$
   $t$-intercepts: $(-2, 0)$, $(0, 0)$




10. domain: $(-\infty, \infty)$
    range: $[-3, 3]$
    $y$-intercept: $(0, -3)$
    $t$-intercept: $\left(\frac{3}{2}, 0\right) = (1.5, 0)$



11. (a)

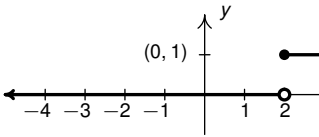



$y = U(t)$

(b) domain: $(-\infty, \infty)$, range: $\{0, 1\}$

(c) $U$ is constant on $(-\infty, 0)$ and $[0, \infty)$.

(d) $U(t - 2) = \begin{cases} 0 & \text{if } t < 2, \\ 1 & \text{if } t \geq 2. \end{cases}$




$y = U(t - 2)$

12. $f(x) = -3$

13. $F(t) = \begin{cases} 2 & \text{if } t \leq 1, \\ -3 & \text{if } 1 < t \leq 3, \\ 4 & \text{if } t > 3. \end{cases}$

14. $L(x) = -\frac{3}{5}x + 1$

15. $g(v) = \begin{cases} 3v + 5 & \text{if } -3 \leq v < -1, \\ 2 & \text{if } -1 < v \leq 3, \end{cases}$

16. (a) $C(20) = 300$. It costs \$300 for 20 copies of the book.

(b) $C(50) = 675$, \$675. $C(51) = 612$, \$612.

(c) 56 books.

17. (a) $S(10) = 17.5$, \$17.50.

(b) There is free shipping on orders of 15 or more comic books.

18. (a) $C(750) = 25$, \$25.

(b) $C(1200) = 45$, \$45.

(c) It costs \$25 for up to 1000 minutes and 10 cents per minute for each minute over 1000 minutes.

19. $d(t) = 3t$, $t \geq 0$.
20. $E(t) = 360t$, $t \geq 0$.
21. $C(x) = 45x + 20$, $x \geq 0$.
22. $C(t) = 80t + 50$, $0 \leq t \leq 8$.

23. $W(x) = 200 + .05x$, $x \geq 0$ She must make \$5500 in weekly sales.

24. $C(p) = 0.035p + 1.5$ The slope 0.035 means it costs 3.5¢ per page. $C(0) = 1.5$ means there is a fixed, or start-up, cost of \$1.50 to make each book.

25. $F(m) = 2.25m + 2.05$ The slope 2.25 means it costs an additional \$2.25 for each mile beyond the first 0.2 miles. $F(0) = 2.05$, so according to the model, it

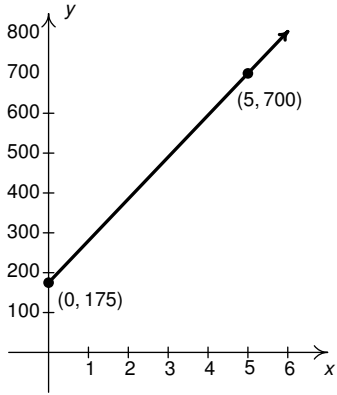
would cost \$2.05 for a trip of 0 miles. Would this ever really happen? Depends on the driver and the passenger, we suppose.

26. (a) $F(T) = \frac{9}{5}T + 32$
    (b) $C(T) = \frac{5}{9}(T - 32) = \frac{5}{9}T - \frac{160}{9}$
    (c) $F(-40) = -40 = C(-40)$.

27. $N(T) = -\frac{2}{15}T + \frac{43}{3}$ and $N(20) = \frac{35}{3} \approx 12$ howls per hour.

    Having a negative number of howls makes no sense and since $N(107.5) = 0$ we can put an upper bound of $107.5^\circ F$ on the domain. The lower bound is trickier because there's nothing other than common sense to go on. As it gets colder, he howls more often. At some point it will either be so cold that he freezes to death or he's howling non-stop. So we're going to say that he can withstand temperatures no lower than $-42^\circ F$ so that the applied domain is $[-42, 107.5]$.

28. (a) $C(0) = 175$, so our start-up costs are \$175. $C(5) = 700$, so to produce 5 systems, it costs \$700.



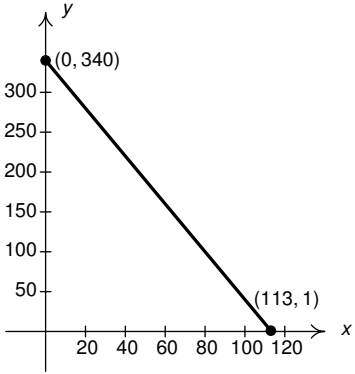

$y = C(x)$

(b) Since we can't make a negative number of game systems, $x \geq 0$.

(c) The slope is $m = 105$ so for each additional system produced, it costs an additional \$105.

(d) Solving $C(x) = 15000$ gives $x \approx 141.19$ so 141 can be produced for \$15,000.

29. (a) $p(x) = -3x + 340$, $0 \leq x \leq 113$.




$y = p(x)$

(b) The slope is $m = -3$ so for each \$3 drop in price, we sell one additional game system.

(c) Since $x = 150$ is not in the domain of $p$, $p(150)$ is not defined. (In other words, under these conditions, it is impossible to sell 150 game systems.)

(d) Solving $p(x) = 150$ gives $x \approx 63.33$ so if the price \$150 per system, we would sell 63 systems.

30. $C(p) = \begin{cases} 6p + 1.5 & \text{if} \quad 1 \leq p \leq 5 \\ 5.5p & \text{if} \quad p \geq 6 \end{cases}$

31. $T(n) = \begin{cases} 15n & \text{if} \quad 1 \le n \le 9 \\ 12.5n & \text{if} \quad n \ge 10 \end{cases}$

32. $C(m) = \begin{cases} 10 & \text{if} \quad 0 \le m \le 500 \\ 10 + 0.15(m - 500) & \text{if} \quad m > 500 \end{cases}$

33. $P(c) = \begin{cases} 0.12c & \text{if} \quad 1 \le c \le 100 \\ 12 + 0.1(c - 100) & \text{if} \quad c > 100 \end{cases}$

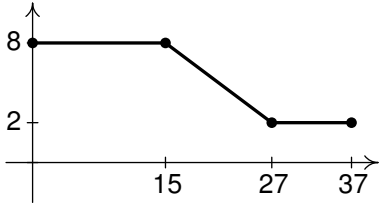
34. (a)

$$D(d) = \begin{cases} 8 & \text{if} \quad 0 \le d \le 15 \\ -\frac{1}{2}\,d + \frac{31}{2} & \text{if} \quad 15 \le d \le 27 \\ 2 & \text{if} \quad 27 \le d \le 37 \end{cases}$$

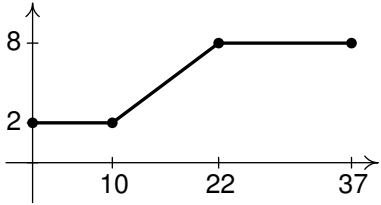
(b)

$$D(s) = \begin{cases} 2 & \text{if} \quad 0 \le s \le 10 \\ \frac{1}{2}\,s - 3 & \text{if} \quad 10 \le s \le 22 \\ 8 & \text{if} \quad 22 \le s \le 37 \end{cases}$$

(c)




$y = D(d)$




$y = D(s)$

35. Since $I(x) = x$ for all real numbers $x$, the function $I$ doesn't change the 'identity' of the input at all.

36. If a graph contains more than one $y$-intercept, it would violate the Vertical Line Test since $x = 0$ would be matched with (at least) two different $y$-values.

37. Vertical Lines fail the Vertical Line Test.

38. $\left(-\frac{b}{m}, 0\right)$. (Note the importance here of $m \neq 0$.)

39. Plugging in $(c, 0)$ for $(x_0, f(x_0))$, we get $f(x) = f(x_0) + m(x - x_0) = 0 + m(x - c)$ or $f(x) = m(x - c)$.

40. Since $L$ is linear with slope 3, $L(x) = L(x_0) + m\Delta x = L(100) + (3)(120 - 100) = L(100) + 60$.

41. (a) $m = \frac{75-64}{4-0} = 2.75$
    (b) $m = \frac{83-75}{8-4} = 2$ (c) $m = \frac{83-83}{10-8} = 0$
    (d) $m = \frac{82-83}{12-10} = -0.5$

    The first two points contributed to a regression line slope of $m = 2.55$; the last two points contributed to a regression line slope of $m = -0.25$.

42. (a) $y = 936.31x - 1645322.6$ with $r = 0.9696$ which indicates a good fit. The slope 936.31 indicates Lake County's population is increasing at a rate of (approximately) 936 people per year.

    (b) According to the model, the population in 2010 will be 236,660.

    (c) According to the model, the population of Lake County will reach 250,000 sometime between 2024 and 2025.

43. (a) $y = 796.8x - 1309762.5$ with $r = 0.8916$ which indicates a reasonable fit. The slope 796.8 indicates Lorain County's population is increasing at a rate of (approximately) 797 people per year.

    (b) According to the model, the population in 2010 will be 291,805.

(c) According to the model, the population of Lake County will reach 325,000 sometime between 2051 and 2052.

44. The regression line is $y = 36.8x + 16.39$ with $r = .99987$, so this is an excellent fit. The slope 36.8 represents mileage in miles per gallon.

45. (c) $y = 0.266x - 459.86$ with $r = 0.9607$ which indicates a good fit. The slope 0.266 indicates the country's energy production is increasing at a rate of 0.266 Quad per year.

(d) According to the model, the production in 2010 will be 74.8 Quad.

(e) According to the model, the production will reach 100 Quad in the year 2105.

46. $\frac{2^3-(-1)^3}{2-(-1)} = 3$

47. $\frac{\frac{1}{5}-\frac{1}{1}}{5-1} = -\frac{1}{5}$

48. $\frac{\sqrt{16}-\sqrt{0}}{16-0} = \frac{1}{4}$

49. $\frac{3^2-(-3)^2}{3-(-3)} = 0$

50. $\frac{\frac{7+4}{7-3} - \frac{5+4}{5-3}}{7-5} = -\frac{7}{8}$

51. $\frac{(3(2)^2 + 2(2) - 7) - (3(-4)^2 + 2(-4) - 7)}{2-(-4)} = -4$

52. The average rate of change is $\frac{h(2)-h(0)}{2-0} = -32$. During the first two seconds after it is dropped, the object has fallen at an average rate of 32 feet per second.

53. The average rate of change is $\frac{F(28)-F(0)}{28-0} = 0.2372$. From 1980 to 2008, the average fuel economy of passenger cars in the US increased, on average, at a rate of 0.2372 miles per gallon per year.

54. (a) $T(4) = 56$, so at 10 AM (4 hours after 6 AM), it is 56°F. $T(8) = 64$, so at 2 PM (8 hours after 6 AM), it is 64°F. $T(12) = 56$, so at 6 PM (12 hours after 6 AM), it is 56°F.

    (b) The average rate of change is $\frac{T(8)-T(4)}{8-4} = 2$. Between 10 AM and 2 PM, the temperature increases, on average, at a rate of 2°F per hour.

    (c) The average rate of change is $\frac{T(12)-T(8)}{12-8} = -2$. Between 2 PM and 6 PM, the temperature decreases, on average, at a rate of 2°F per hour.

    (d) The average rate of change is $\frac{T(12)-T(4)}{12-4} = 0$. Between 10 AM and 6 PM, the temperature, on average, remains constant.

55. The average rate of change is $\frac{C(5)-C(3)}{5-3} = -2$. As production is increased from 3000 to 5000 pens, the cost decreases at an average rate of \$200 per 1000 pens produced (20¢ per pen.)

56. (a) i. $-49.5$ so the average velocity of the rocket between 14.9 and 15 seconds after lift off

is $-49.5$ feet per second (49.5 feet per second directed *downwards*.)

ii. $-50.5$ so the average velocity of the rocket between 14 and 15.1 seconds after lift off is $-50.5$ feet per second. (50.5 feet per second directed *downwards*.)

iii. $-49.95$ so the average velocity of the rocket between 14.99 and 15 seconds after lift off is $-49.95$ feet per second. (49.95 feet per second directed *downwards*.)

iv. $-50.05$ so the average velocity of the rocket between 15.01 and 15 seconds after lift off is $-50.05$ feet per second. (50.05 feet per second directed *downwards*.)

(b) The average rate of change seem to be approaching $-50$.

(c) Line: $y = -50(t - 15) + 375$ or $y = -50t + 1125$. Graphing this line along with the $s$ on a graphing utility we find the two graphs become indistinguishable as we zoom in near $(15, 375)$.

60. (a) i. $L(x) = 3$ ii. $L(x) = -2$
iii. $L(x) = x + 1$ iv. $L(x) = -2x + 3$

## 1.3 Absolute Value Functions

## 1.4 Quadratic Functions